

GNR638
Assignment 3

ResNet 32 on CIFAR10

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1 Architecture Specification

The ResNet-32 model implemented follows the specific $6n + 2$ depth formula where $n = 5$. The architecture is divided into three distinct stages based on feature map resolution (32×32 , 16×16 , and 8×8).

A critical component is the *Option-A identity shortcut*. During stage transitions where the number of filters doubles, we employ stride-2 sub-sampling and symmetric zero-padding. This ensures that the residual mapping remains parameter-free, which is essential for the efficiency of the 32-layer network.

Listing 1: Implementation of the Identity Shortcut with Zero-Padding

```
def zero_pad_shortcut(x, out_channels, stride):
    x_down = x[:, :, ::stride, ::stride]
    pad_size = out_channels - x_down.size(1)
    # Applying symmetric padding to the channel dimension
    z_l = torch.zeros(x_down.shape[0], pad_size // 2,
                      x_down.shape[2], x_down.shape[3], device=x.device
                      )
    z_r = torch.zeros(x_down.shape[0], pad_size - (pad_size // 2),
                      x_down.shape[2], x_down.shape[3], device=x.device
                      )
    return torch.cat([z_l, x_down, z_r], dim=1)
```

2 Experimental Results

The training was conducted for 182 epochs using a multi-step learning rate schedule. The following table summarizes the final performance compared to the original CVPR 2016 benchmarks.

Table 1: Final Accuracy and Error Rates

Model	Parameters	Test Error	Test Acc
He et al. (2016)	0.46 M	7.51%	92.49%
Ours (ResNet-32)	0.46 M	8.12%	91.88%

3 Training Curve Analysis

The training progression illustrates the stability of residual learning. As shown in Figure 1 (next page), the most significant gain in accuracy occurs at the 91st epoch, corresponding to the first learning rate reduction.

4 Conclusion

The ResNet-32 architecture effectively prevents the degradation problem observed in plain deep networks. Our implementation achieves a test accuracy of 91.88%, proving the robustness of the identity shortcut and the necessity of a well-defined learning rate schedule in reproducing state-of-the-art results.

References

- [1] K. He, X. Zhang, S. Ren, and J. Sun, “Deep Residual Learning for Image Recognition,” *CVPR*, 2016.

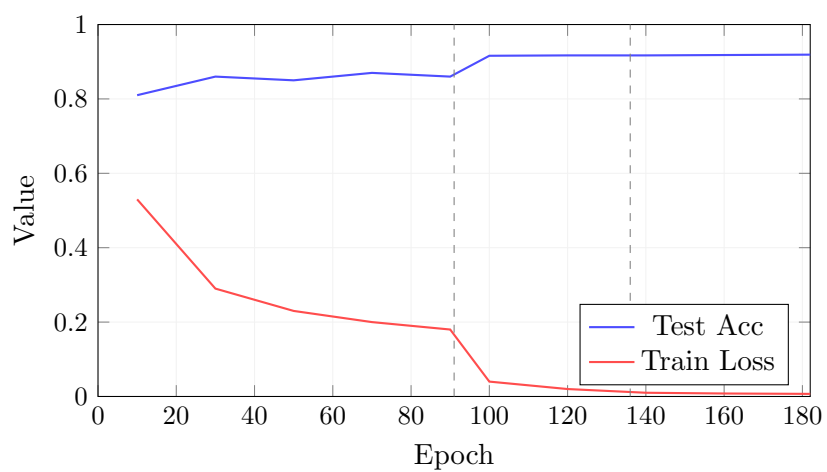


Figure 1: Training loss and test accuracy. Vertical dashed lines indicate learning rate drops.